Entries to Networks – Transforming Biographical Dictionaries into Knowledge Graphs with LLMs

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Abstract

Semi-structured historical sources like biographical dictionaries or lexicons constitute rich but often underutilized resources for Historical Network Research. While containing extensive relational information with high density, they remain largely disconnected internally, accessible mostly via sequential reading of individual entries. Large Language Models (LLMs) offer new possibilities for extracting structured information like semantic triples from such sources at scale, enabling the construction of graph structures that support advanced analytical tasks, including network analysis. Despite these advances, significant challenges remain, particularly in handling specialized language or symbol systems, disambiguation, validation of extracted (or inferred) relations, and the opacity of LLM outputs. We address these challenges by developing a modular, two-stage pipeline incorporating multiple validation steps and human-in-the-loop supervision (Schlattmann et al. 2026). An ontology-agnostic Open Information Extraction (OIE) stage generates candidate relations, while a second, research-question-driven ontology stage guides final structuring and validation. We apply this pipeline to two cases. First, we study effects of nineteenth- and early twentieth-century Polish migration by extracting persons, publications, and references from *A Little Dictionary of Polish Colonial and Maritime Pioneers* and related bibliographic sources (Zieliński 1933, 1935), asking which actors and publication venues functioned as hubs in transnational knowledge circulation processes and how these patterns shift over time (1830–1930). Second, we examine the interconnected development of East German scientific and political elites using *Who Was Who in the GDR?* (Müller-Enbergs et al. 2010). These sources share a similar structure while differing in language, period, and historical context. Networks constructed from these extractions reveal structural patterns previously hidden in isolated entries, enabling the combination of micro-level biographical detail with macro-level network perspectives. The pipeline also exposes source-inherent biases that only become visible via structural analysis. We discuss practical challenges, such as historical disambiguation, and the epistemic caution required when integrating LLMs into historical research.

References

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